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articles

New palaeomagnetic evidence fails to support rotation of western Newfoundland

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Palaeomagnetic results from Ordovician to late Precambrian rocks in western Newfoundland are inconsistent with Wegener's hypothesis that the island has rotated 30° anticlockwise. Previous evidence apparently supporting this hypothesis is shown to be inconclusive.

WEGENER[1] proposed that Newfoundland has rotated 30° anticlockwise away from Labrador, implying a nearby pivot. A palaeomagnetic test of this motion is, in principle, straightforward, provided that test sites are carefully chosen (Fig. 1). Even so, the resolution obtainable may be small or zero, being a function of both the rotation angle R and the angular distance θ between pivot and ancient pole. Figure 2 illustrates this for Newfoundland ($R=30°$), where values of P ('resolution') approaching 30° are obtainable at Lower Palaeozoic sites suitable for testing possible Devonian or later rotations. Hence a near-optimum test is feasible because of the low latitude of ancient Newfoundland.

After an inconclusive application of this test[2,3], Black[4] compared Palaeozoic rocks of matching age in western Newfoundland and mainland eastern Canada, concluding that his data uphold a 30° anticlockwise rotation of at least western Newfoundland during the middle to late Devonian. Black argued that this would explain observed regional structures in the north-eastern Appalachians. Except for differences in latitude, his widely quoted Cambrian and Devonian comparisons conform strikingly to the positive test in Fig. 1. Regional and local tectonics[5,6] and seismic results from the Gulf of St Lawrence[7] have been cited as evidence against this rotation. Further, Black's data were found inconclusive[8], because (1) the Cambrian remanence directions inferred in Newfoundland and New Brunswick (Fig. 3, S, R) are not significantly different; (2) the respective "Devonian" directions may not be contemporaneous; (3) by demagnetising only in alternating fields (AF) to 300 Oe, he was unlikely to have distinguished different high-stability components. Also[9], since the mainland sites (Fig. 3, R) form part[10] of the Avalon tectono-stratigraphic zone, rocks on opposite sides of the Central Mobile Belt were being compared (R, S). Therefore data in Fig. 1 could equally be taken to support either a rotation of Newfoundland or a proto-Atlantic model[11].

New magnetic data

To meet these objections we undertook new magnetic studies of mid-Ordovician to latest Precambrian rocks in Newfoundland (Fig. 3, N1–N3, N5–N7), using standard equipment. The results, summarised below and in Table 1, will be reported in greater detail elsewhere. Comparisons are with published data from tectonically stable mainland regions (Table 2).

Useful palaeomagnetic results have been obtained from carbonate rocks[12–14]. We collected limestone of the Lower Ordovician[15,16] St George Formation at 10 sites (62 samples) in the Port-au-Port area (N1). The natural remanence (NRM), carried in different samples by magnetite, haematite or both (compare ref. 14), was weak and fewer than half of the samples could be demagnetised. Haematite samples were unaffected by AF treatment (800 Oe) while thermal treatment was inconclusive, but in the magnetite samples both methods produced comparable, better-grouped new remanence directions (Table 1). St George limestone from Cape Norman (N2) gave similar results. This magnetic

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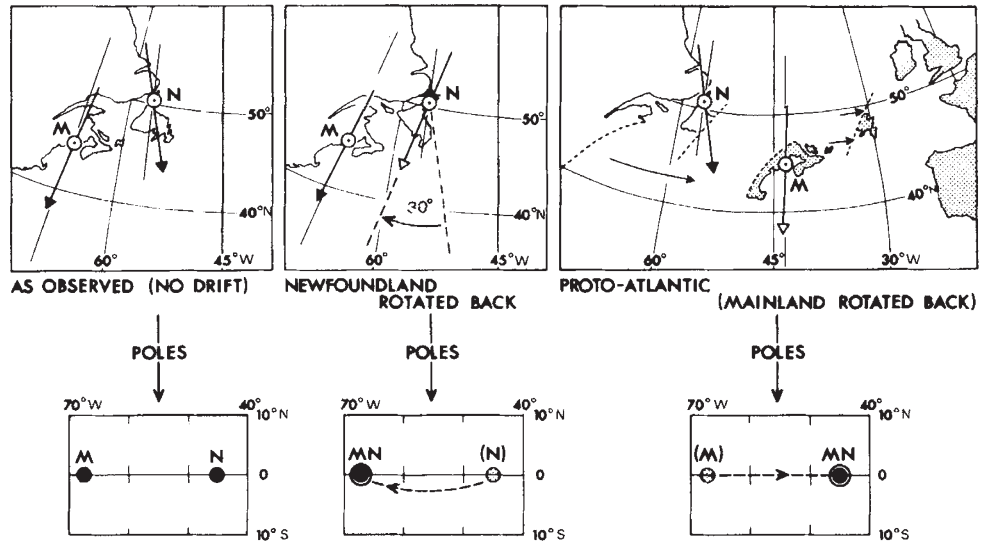


Fig. 1 Palaeomagnetic test of rotation (schematic). Declinations of hypothetical 'observed' remanence vectors (solid arrows) and corresponding pole positions are shown for Newfoundland (N) and the adjacent mainland (M). Hollow arrows depict the restored vectors when either Newfoundland is rotated back[1] or the mainland site is rotated to a presumed location before the closing of a proto-Atlantic ocean[11]. The pivots are at site N and 90°N, respectively. For convenience, poles have been located at the present equator. The two rotations here are indistinguishable[9]: an unambiguous test of the Newfoundland rotation requires comparison with a mainland site known not to have rotated. Stippled portions (right) correspond to the Atlantic faunal realm[11].

behaviour in Newfoundland limestones parallels that found earlier in the mid-Ordovician Trenton limestone[12] (M1), where it was shown that the stable component originated before the completion of diagenesis. This suggests a similar origin in the Port-au-Port limestone, but proof is difficult and it is uncertain whether one should correct for the gentle (12–15°) tilt which is associated[10] with the Taconic (pre-mid-Ordovician) or Acadian (Devonian) orogeny or both. The (uncorrected) mean directions for areas N1 and N2 after AF or thermal treatment are all similar (Table 1). Pole N1 (Table 2) is based on sample rather than site statistics, since too few samples per site remained measurable after AF treatment. The tilt-corrected pole is 22°N, 120°E. Pole N1 is not significantly different from Pole N4 (ref. 13) for the St George Formation, where no tilt correction was reported.

A result[17] from pillow basalt and sheeted dikes of the Moreton's Harbour Group (N3) is useful for comparison with the St George Formation, since different rock types of similar ages are being compared. Most or all of the 8-km thick Moreton's Harbour sequence has been attributed[18,19] to island arc volcanism in the Lower[19] to Middle[18] Ordovician. Although these rocks lie in the Central Mobile Belt (Fig. 3), it seemed reasonable to include them in a test based on western Newfoundland, since they probably originated as part of the North American margin (D. F. Strong, personal communication). A stable remanence in magnetite, isolated by AF demagnetisation, presumably was acquired not later than the generally low-grade greenschist metamorphism of these rocks. The tilt-corrected pole (Table 2) is not significantly different from poles N1 and N4 (Fig. 4).

Lower Cambrian[15] Bradore sandstones sampled by Black[4] at St John Bay (Fig. 3, S) were re-sampled more extensively[20] at Canada Bay (N5). The NRM, carried by haematite, had a mean direction as in Black's AF-treated samples (Table 1), but by thermal demagnetisation a stable component misaligned with both the NRM and Black's direction was isolated at 665 °C: this shows that the objection[8] to exclusive low-AF treatment is well-founded. This component probably pre-dates the gentle (15–25°) Taconic and/or Acadian tilt of the sediments, though a post-tilting origin of the entire NRM is not impossible[20]. The stable direction corresponds to a reverse pole at 23°N, 160°E

Table 1 New palaeomagnetic results from Newfoundland

Area no.	Rock unit/area	Geological age	N	(n)*	Mean NRM direction† Declination (°)	Inclination (°)	k	Treatment‡	Mean direction after treatment† Declination (°)	Inclination (°)	k	α_{95}(°)
N1	St George limestone, Port-au-Port area	Early Ordovician[15]	5S§	(14s)	179.5	+62	63	300 Oe	181.5	+25	910	2.5
			15s§	—	179.5	+57.5	13	300 Oe	182	+23	90	4
			6s	—	159	+61.5	49	500 °C	183	+18.5	200	5
N2	St George limestone, Cape Norman area	Early Ordovician[15,16]	5s	—	164.5	+73	70	400 °C, ‖ 300 Oe ‖	177	+22.5	120	7
N3	Moreton's Harbour basalt	Early to mid-Ordovician[18,19]	11S	(56s)¶	114	+66	8	150 Oe	(116.5 176	+57 +16	9 9	16.5) 16
N5	Bradore sandstone, Canada Bay	Early Cambrian[15]	9S	(54s)	153.5	+52	13	665 °C	146.5	+18	110	5
N6	Bradore sandstone, St John Bay	Early Cambrian[15]	2S	(13s)	198	+66	—	665 °C	132.5	+25.5	—	—
S	Bradore sandstone, (Black's result)[4]	Early Cambrian[15]	10s	—	?	?	?	300 Oe	151	+44.5	27	9.5
N7	Cloud Mountain basalt	Latest Precambrian**	3F	(10S)	159.5	+73	14	100 Oe	(142 127	+50 +36.5	840 1250	4.5°) 3.5°

*N, number of flows (F), sites (S) or samples (s) used in averaging; (*n*) number of sites or samples studied.
†Angles rounded to 0.5°; *k* values greater than 100 are rounded to nearest 10; where two sets of data are given, those in brackets were obtained before correction for geological tilt and those without brackets are tilt-corrected and were used in Table 2. In all other cases, no tilt correction was made.
‡Stepwise thermal or AF demagnetisation, with fields quoted in peak oersteds.
§Site and sample statistics for the same samples are shown for comparison; samples (second row) are used in Table 2.
‖Three samples were treated to 400 °C, two samples to 300 Oe.
¶Reduced to 46 samples (11 sites) after treatment.
**$^{40}Ar/^{39}Ar$ date of 605±10 Myr cited for Long Range dikes (ref. 25), which are considered (refs 24, 25) to be contemporaneous with Cloud Mountain basalts.

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Table 2 Palaeomagnetic poles for Newfoundland and mainland North America

Area no.	Rock unit/area	Geological age*	N†	Treatment†	Pole position Latitude (°)	Longitude (°)	dp,dm (°)	Polarity†	Mean rotated pole‡ Latitude (°)	Longitude (°)
N1	St George limestone, Port-au-Port area	Early Ordovician	15s	a	30 N	119 E	2,3	R	26 N	86 E
N3	Moreton's Harbour basalt	Early to mid-Ordovician	11S	a	32 N	130 E	9,17	R	27 N	96 E
N4	St George dolostone, Daniel's Harbour	Early Ordovician[15,13]	8S	a	29 N	125 E	3,7§	N	25 N	92 E
M1	Trenton, limestone, New York	Mid-Ordovician[12]	29s	a	36 N	114 E	3,5	R		
M2	Front Range intrusives, Colorado	'Cambro-Ordovician'[26]	18s	a	37 N	122 E	6,12	R		
			14s	t	44 N	100 E	5,11	R		
			—	a+t	40 N	111 E	—	R		
N5	Bradore sandstone, Canada Bay	Early Cambrian[15]	9S	t	23 N	160 E	3,5	R	30 N	128 E
N7	Cloud Mountain basalt	Latest Precambrian	3F	a	5 N	172 E	2,4	R	17 N	146 E
M3	Franklin intrusives, Northwest Territories	Latest Precambrian[29,31]	46S	a	6 N	166 E	4,4‖	M		
N8	Indian Head anorthosite	'Grenville'[32]	5S	a	8S	158 E	15,20	N	1 N	136 E
M4	Morin anorthosite, Quebec	'Grenville'[33]	13S	a	O	164 E	9,12	R		
M5	Northern Grenville anorthosite, Ontario	'Grenville'[34]	4S	a	8 N	161 E	6,6‖	R		
M6	Whitestone anorthosite, Ontario	'Grenville'[35]	8S	a	5 S	168 E	6,10	R		
M7	Charlevoix anorthosite, Quebec	'Grenville'[36]	5 S	a	7 S	148 E	12,16	R		

*Ages are discussed in the references (see also Table 1). Ref. 26: Radiometric ages in the range 601–427 Myr are cited. Ref. 29: The authors cite numerous K/Ar ages with 'best mean age' 650 Myr, but they adopt 675 Myr, to allow for argon loss. Ref. 31: Other Franklin igneous sites have K/Ar ages matching an isochron of 625 Myr. Refs 32–36: 'Grenville' ages in the range $0.9–1.0 \times 10^9$ yr are cited, based on radiometric data or on inferred time of magnetisation.

†*N*, see footnote to Table 1; Treatment: a, alternating-field; t, thermal; a+t, mean of poles based on a and t treatment; Polarity: N, normal; R, reverse; M, mixed.

‡Average values are quoted for 30° clockwise rotations about pivots A (51.5°N, 56.5°W) and B (48°N, 59.5°W) (Fig. 3).

§Recalculated from original data.

‖Error circles (α_{95}) instead of ovals are cited by authors.

(Table 2), or 29°N, 167°E after tilt correction. Bradore standstones from St John Bay (N6) gave comparable (virtual) poles.

The gently dipping Cloud Mountain plateau basalts (N7) were sampled at 10 sites. They have been correlated stratigraphically[21] and through magnetic similarities[22,23] with plateau basalts in south-east Labrador of presumed late Precambrian to early Cambrian age[21]. Strong and Williams[24] consider the Cloud Mountain basalts to be co-magmatic with and contemporaneous with the chemically similar Long Rang dikes, which have a $^{40}Ar/^{39}Ar$ dating of 605 ± 10 Myr (ref. 25). Only three separate flows could be identified. AF treatment isolated a stable component carried by magnetite, resulting in greatly improved within- and between-flow vector groupings. Its mean direction after tilt correction corresponds to the pole shown in Table 2. We note that a residual secular variation component may be present in these few flows.

Rotation test

There is no unique pivot for the postulated rotation of western Newfoundland. Black[4] relocated his Newfoundland palaeomagnetic poles by rotating the island 30° clockwise separately about two extreme pivots A and B (Fig. 3) using the mean restored pole position to compare with poles on the mainland. A similar procedure is followed here (Table 2). The uncertainty in pole positions due to averaging about the two pivots is ⩽1°.

Out of the three identical Ordovician Newfoundland poles, N1 and N3 are in turn identical with, and pole N4 is close to, the Trenton limestone pole[12] (M1). Here 'identical' is defined non-rigorously by an overlap of the 95% confidence ovals (Fig. 4). Pole M2 from the Front Range Intrusives[26] of uncertain Cambrian to Ordovician age is shown without oval, being the average of two poles equally weighted in ref. 26, based on AF and thermal treatment, respectively; the former but not the latter pole is identical to poles N1, N3, N4 and M1. With Newfoundland rotated back, the 'N' and 'M' poles become discordant, except pole N3 for which the test is inconclusive. These results are

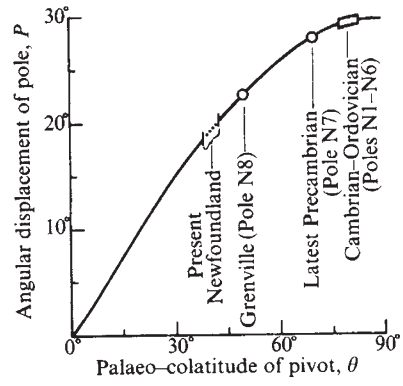


Fig. 2 Resolution (*P*) obtainable by palaeomagnetism for testing rotation (*R*) of a land mass, as a function of the angular distance (θ) between pivot and ancient pole, where[9]: $1-\cos P = \sin^2\theta\,(1-\cos R)$. *P* is the angular shift of the ancient pole on restoring the land mass to its pre-drift position. *P* cannot exceed *R*. Illustrated for the Newfoundland hypothesis ($R = 30°$) using actual results (Table 2). Note that, with minor error in θ, site locations N1–N8 may be substituted for the average pivot locations A and B (Fig. 3).

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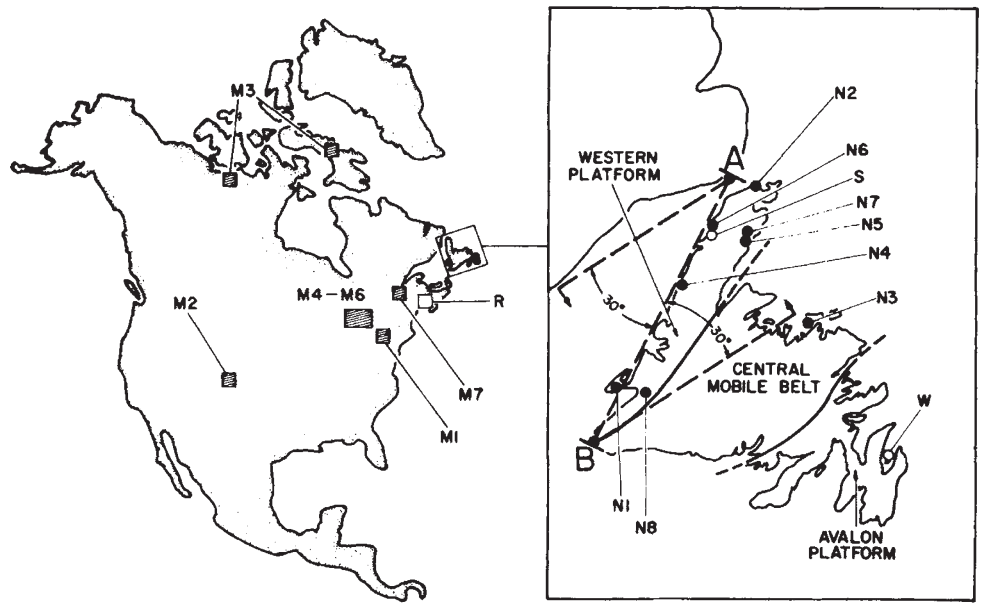


Fig. 3 Palaeomagnetic sites for testing rotation of Newfoundland (Tables 1, 2). R, Ratcliffe Brook, New Brunswick (Lower Cambrian)[4]; M1–M7, other mainland sites. S, St John Bay[4]; W, Wabana[37]; N1–N7, other Newfoundland sites. A, B, alternative pivots[4] (Table 2).

incompatible with the proposed rotation, at least since mid-Ordovician time. A provisional Upper Ordovician pole (R. Van der Voo, personal communication) at 29°N, 118°E from thermally cleaned red sandstones of the Juniata Formation, Pennsylvania, is not significantly different from poles N1 (30°N, 119°E) N3, N4 and M1. This agreement may be due to insignificance of mid- to late-Ordovician polar motion relative to eastern North America, in which case the Juniata result is additional evidence against rotation of western Newfoundland.

Comparisons of the Bradore data[20] (N5) with those from Cambrian mainland localities that avoid the proto-Atlantic ambiguity are difficult, since the few reliable mainland poles (see ref. 27) have been cited as Upper rather than Lower Cambrian in age; also they are widely dispersed, suggesting rapid polar displacement. Hence a rotation test using the Bradore result seems premature. A Lower to Middle Cambrian pole at 13°N, 157°E for eastern Quebec was recently reported[28], however. Its fair agreement with the Bradore pole (23°N, 160°E) is destroyed when Newfoundland is rotated back. This may be taken as evidence against the proposed rotation.

On the mainland the youngest suitable Precambrian rocks are the Franklin intrusives[29–31]. Of numerous K–Ar datings, 675 Myr is the adopted age[30] cited for pole M3 (Table 2). but other Franklin diabases and lavas[31] give a pole close to M3 and are dated 625 Myr, compared with the 605-Myr age inferred for Cloud Mountain (N7). Figure 4 shows agreement between poles N7 and M3, but they strikingly diverge when Newfoundland is rotated back. Pending better knowledge of the Newfoundland–mainland age difference and of the time span averaged in the Cloud Mountain lavas, this result again is inconsistent with the rotation hypothesis.

The minimum radiometric age of 890–900 Myr of the Indian Head anorthosites[32] (N8) suggests that they may be contemporaneous with mainland rock formations dated at the time of the Grenville orogeny (1.0×10^9 yr). Pole N8 was compared with only mainland poles dated $0.9–1.0\times10^9$ yr, based on anorthosite (M4–M7)[33–36], to make it more probable that a synchronous Grenville event is being tested. Once again an initial agreement between 'N' and 'M' poles is lessened when Newfoundland is rotated back (Fig. 4). Unrotated, the oval of pole N8 overlaps all mainland ovals, whereas after rotation it misses two of these (M5, M6). The large size of oval N8, however, makes this test only marginally significant.

Significance of the test

Newfoundland data from the Ordovician, latest Precambrian, more marginally the Grenville and perhaps also the Cambrian, when compared with the mainland results, provide evidence against Wegener's[1] hypothesis. At the

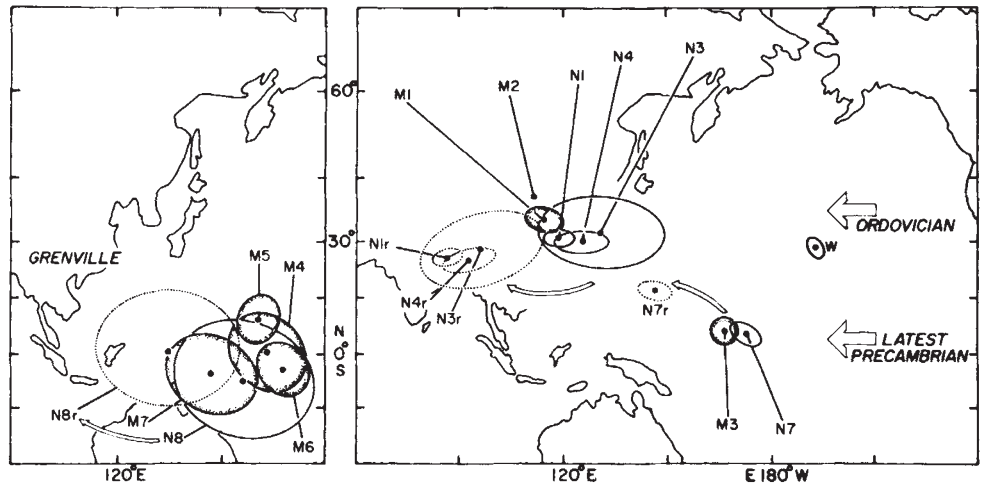


Fig. 4 Negative results of the rotation test for three age levels. Pole positions with 95% confidence ovals (Table 2) for the mainland (M) (shaded ovals), and Newfoundland (N) as observed (solid ovals) and after rotation (dotted ovals, site numbers followed by '*r*'). W, Wabana (Fig. 3 and text)[37]. Magnetic north or south poles are plotted to correspond to presumed north rotation poles.

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same time we have verified that Black's[4] Cambrian results used by him in support of that hypothesis are inconclusive. Generally better data were available to us than to the earlier workers[2,4] in terms of statistics, magnetic control and comparative dating. Limitations are, however—(1) some large confidence ovals; (2) different populations (samples, sites, flows) being compared; (3) the small number of usable limestone samples (15) and of Cloud Mountain flows (three); (4) uncertainty regarding tilt corrections; (5) possible anachronisms between poles of any given age; and (6) the absence of a reliably Lower Cambrian mainland result to compare with pole N5.

Partly because of these limitations, we could not employ rigorous statistics. For example, an F test applied to negative evidence on rotation would ideally show, in each case with high probability, that the mainland and unrotated Newfoundland poles for a certain age are identical and that they become distinct when Newfoundland is rotated back. Comparing actual data, such as Ordovician poles N1, N4, M1, the latter condition is certainly met, but the former condition is only moderately probable. The difficulty is predictable from Fig. 4 and recurs in an F test of the Grenville poles. This restricts us to the use of confidence ovals for assessing significance.

We conclude that the evidence is inconsistent with western Newfoundland having rotated by 30° since the late Precambrian, and is consistent with the absence of any rotation, although a rotation by 5°–10° since the Ordovician is possible. To determine whether such lesser rotation actually took place calls for more refined palaeomagnetic data from Newfoundland and North America.

Finally, a Lower Ordovician pole from Wabana, eastern Newfoundland[37] (W, Figs 3, 4) is seen to be far displaced from poles N1 and N4 of similar age in western Newfoundland. This is compatible with the hypothesis[11] of a Lower Palaeozoic proto-Atlantic Ocean of undetermined width, in which the two portions of Newfoundland occupied opposite coasts. Poles W, N1 and N4 may be made to coincide by rotating Wabana towards Europe about some pivot in the central Atlantic.

We thank R. R. Pätzold, Dr P. S. Rao, H. G. Dawe, T. Vallis and P. Kirby for assistance. This research was supported by the National Research Council of Canada.

Received 13 October 1976; accepted 13 January 1977.

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Current rifting episode in north Iceland

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A major rifting episode is now occurring in north Iceland. This started on 20 December 1975, with a basaltic eruption, an exceptionally intense earthquake swarm and movement on an 80-km segment of the plate boundary. Inflation and deflation of the Krafla caldera indicate upwelling of magma and injection into the rift zone. Historical records show that similar episodic rifting occurs in this region every 100–150 yr.

THE plate boundary between the American and European plates runs through Iceland as a zone of active volcanism, seismicity and graben structures. The two plates are drifting away from the boundary at a velocity of about 1 cm yr^{-1} (ref. 1), but the mechanism of this drifting process at the boundary itself has not been clearly observed until now. A spectacular rifting episode started in northern Iceland on 20 December 1975, and is still not completed in December 1976. Intense earthquakes, widespread ground deformation and magmatism are affecting an 80-km long segment of the plate boundary. We describe here the geological setting of these events, the main observations to date and an attempted interpretation.

In north Iceland the Neovolcanic Zone has a north–south direction. Towards the south it connects with the Eastern Volcanic Zone of south Iceland, but towards the north it is offset to the west by about 100 km, by the Tjörnes Fracture Zone, to join the Kolbeinsey Ridge (Fig. 1). The structure of the Neovolcanic Zone in north Iceland

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